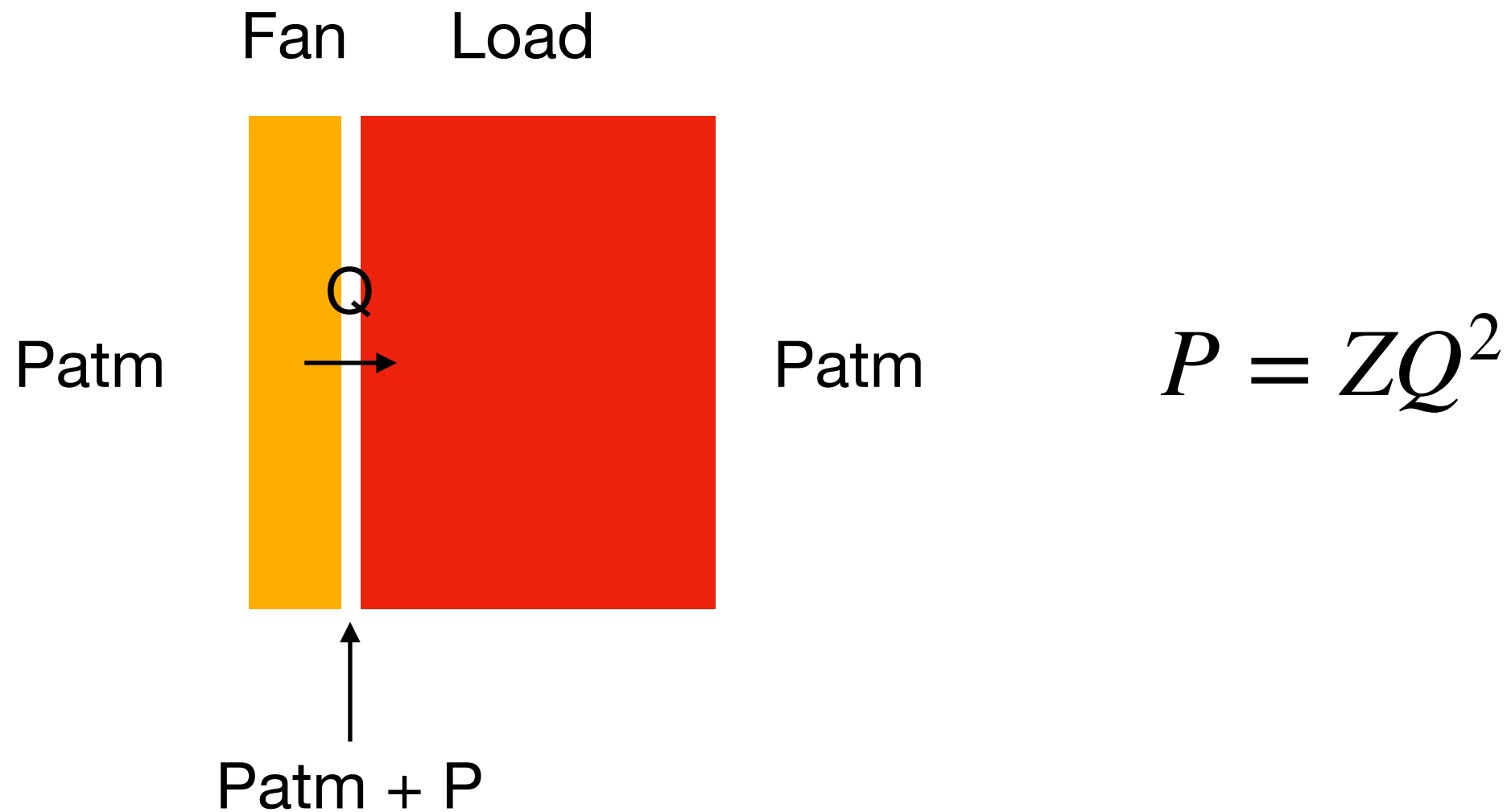


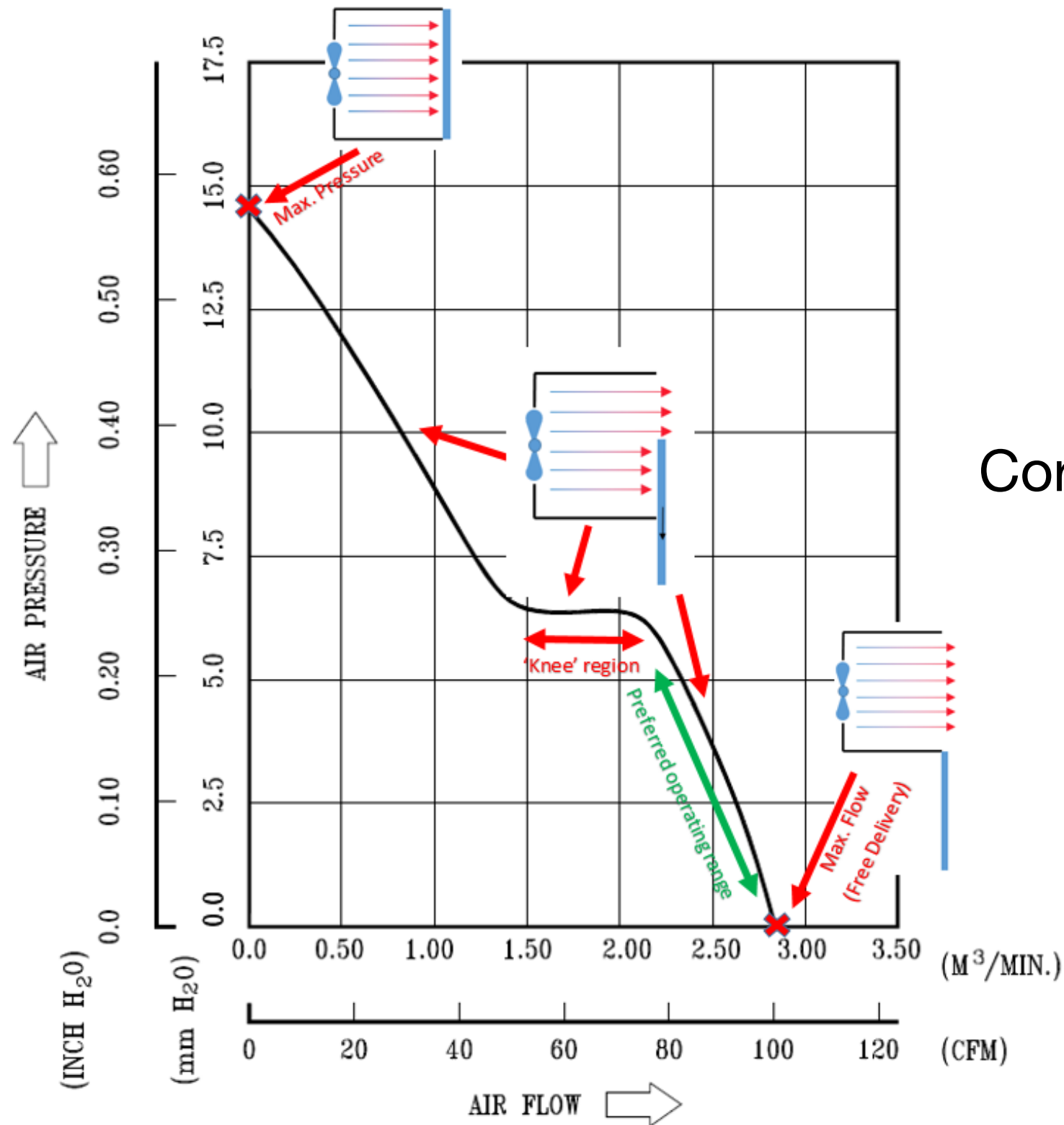
DC fan laws

Definitions and system curve



PQ curve: basics

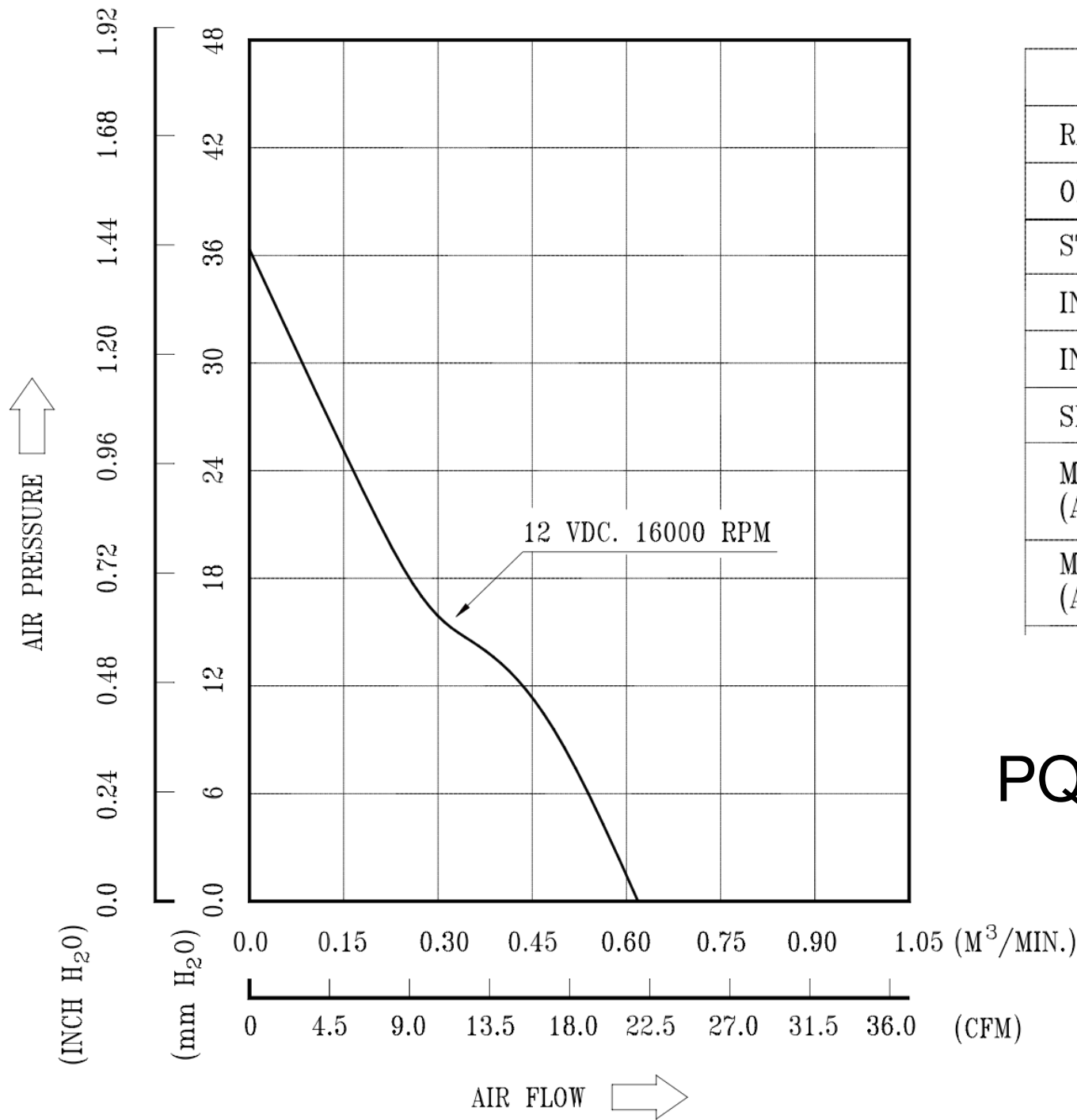
9. P & Q CURVE:



Corresponds to a **variety of loads**
for a **fixed voltage (RPM)**

PQ curve

Data sheet of PFB0412EHN-TP06



ITEM	DESCRIPTION
RATED VOLTAGE	12 VDC
OPERATION VOLTAGE	10.8 – 13.2 VDC
START VOLTAGE (VDC)	≥ 10.8 VDC.
INPUT CURRENT	0.50 (MAX. 0.60) A
INPUT POWER	6.00 (MAX. 7.20) W
SPEED	16000±10% R.P.M.
MAX. AIR FLOW (AT ZERO STATIC PRESSURE)	0.618(MIN. 0.556) M ³ /MIN. 21.834 (MIN. 19.643) CFM
MAX. AIR PRESSURE (AT ZERO AIRFLOW)	36.334(MIN. 29.430) mmH ₂ O 1.431 (MIN. 1.159) inchH ₂ O

PQ curve at rated voltage and speed

Affinity laws

When modifying the applied voltage

Fan and air speed vary **linearly**

$$\frac{\text{RPM}_1}{\text{RPM}_0} = \frac{Q_1}{Q_0} = \frac{V_1}{V_0}$$

Pressure varies **quadratically**

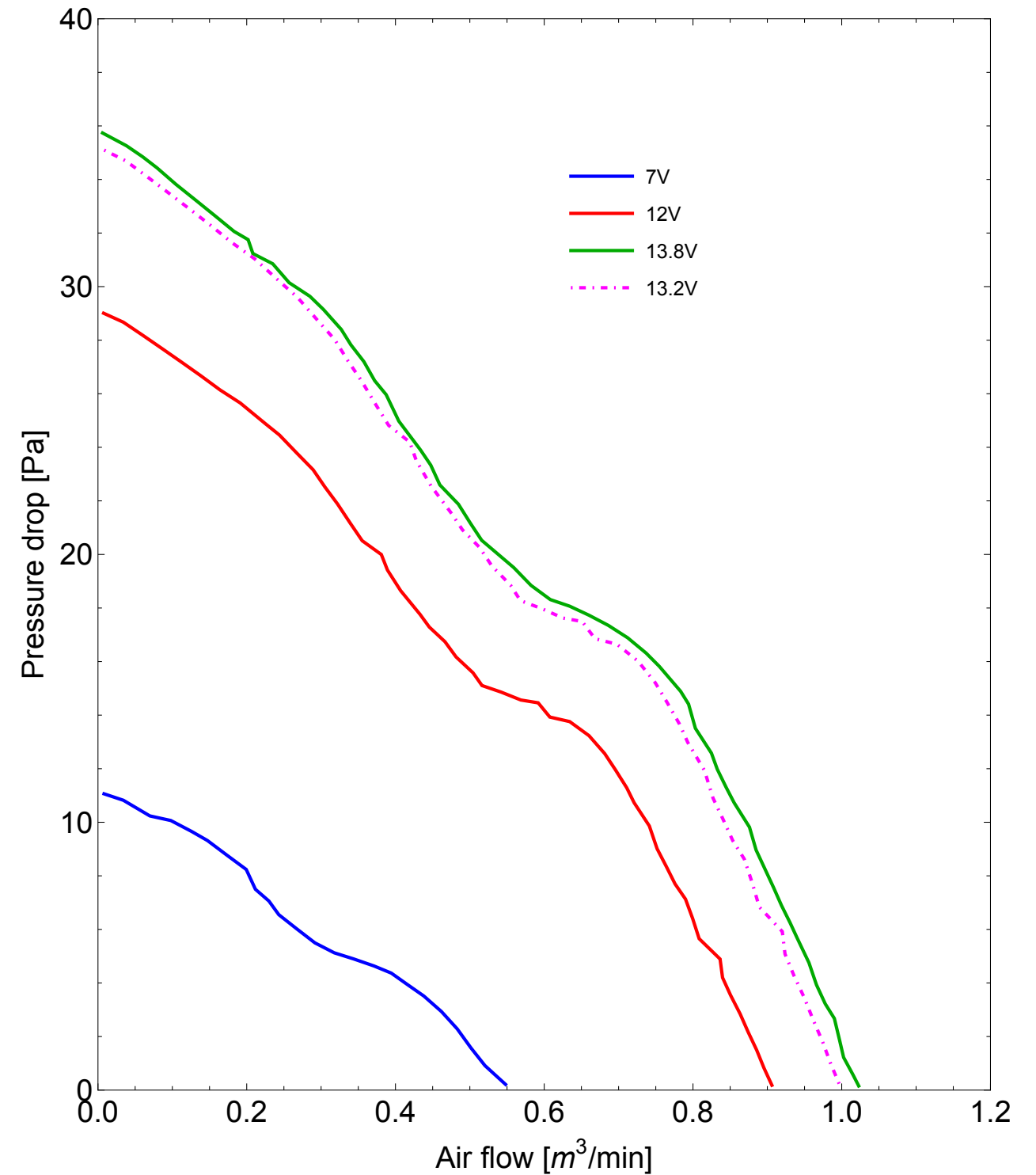
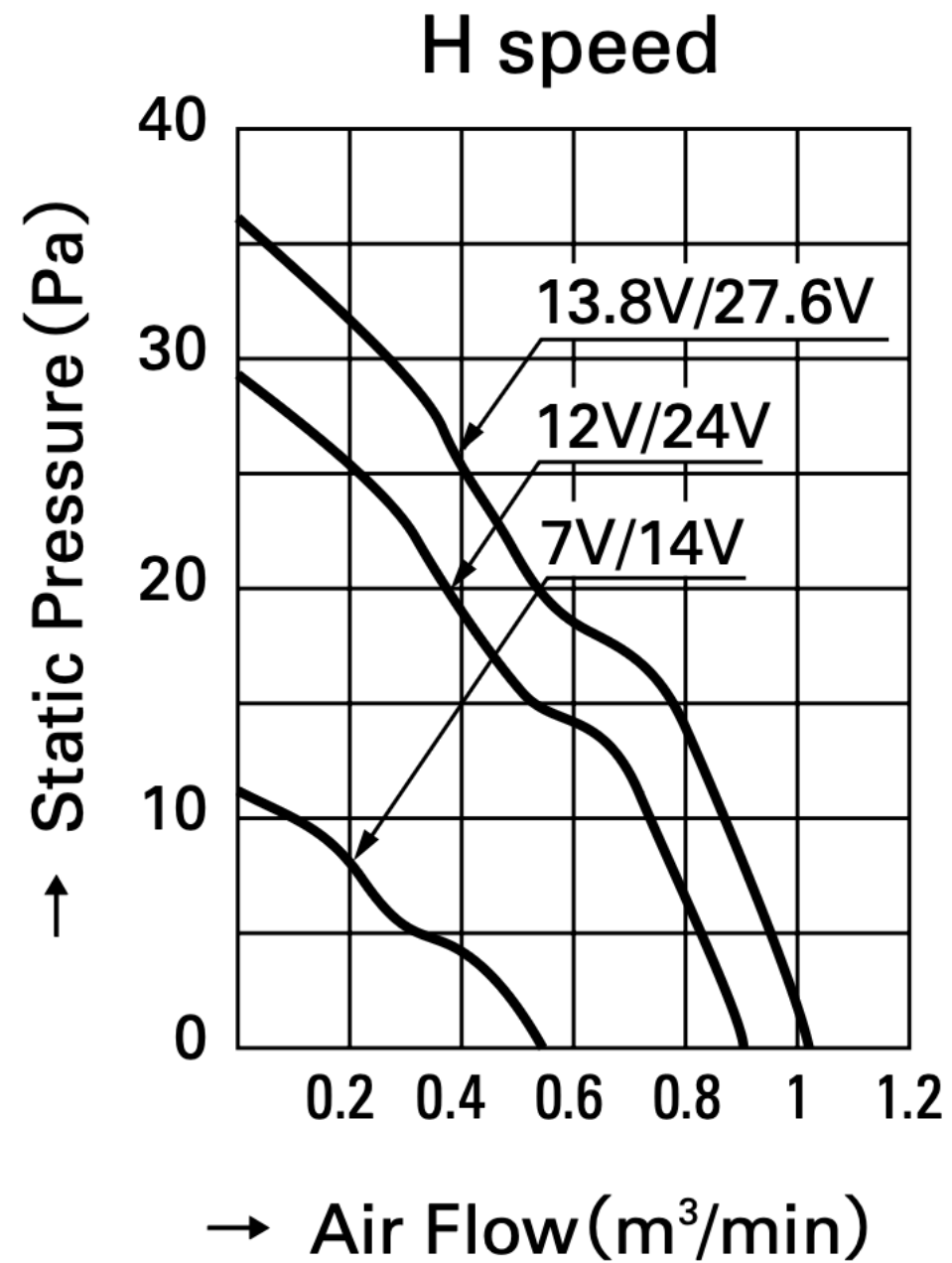
$$\frac{P_1}{P_0} = \left(\frac{V_1}{V_0} \right)^2$$

Dissipated power varies **cubically**

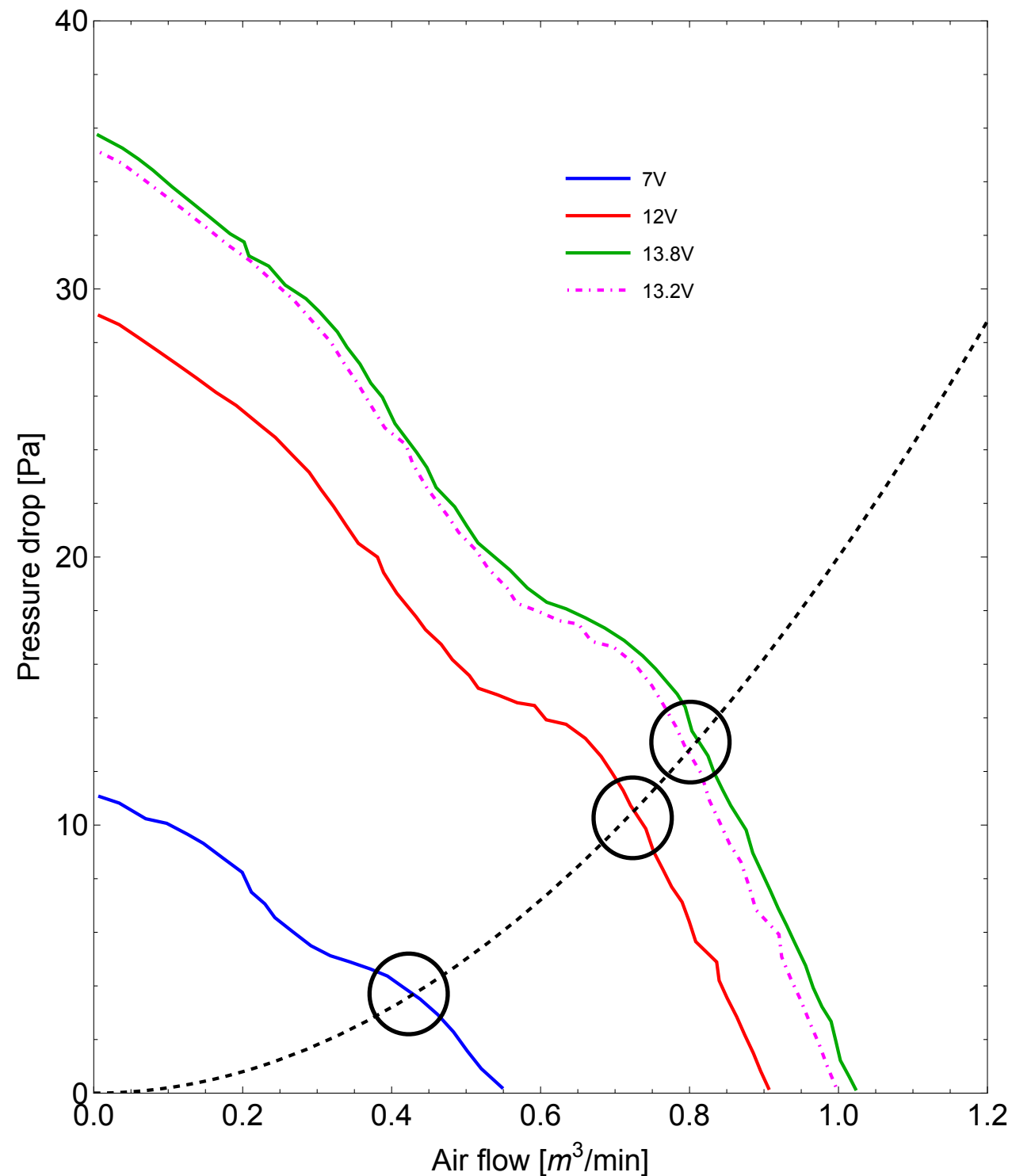
$$\frac{\text{HP}_1}{\text{HP}_0} = \left(\frac{V_1}{V_0} \right)^3$$

Example on a data sheet

Data sheet of 109P0812H701



System curve and operating point



System curve

$$P = ZQ^2$$

Operating point

$Q(V)$ $P(V)$

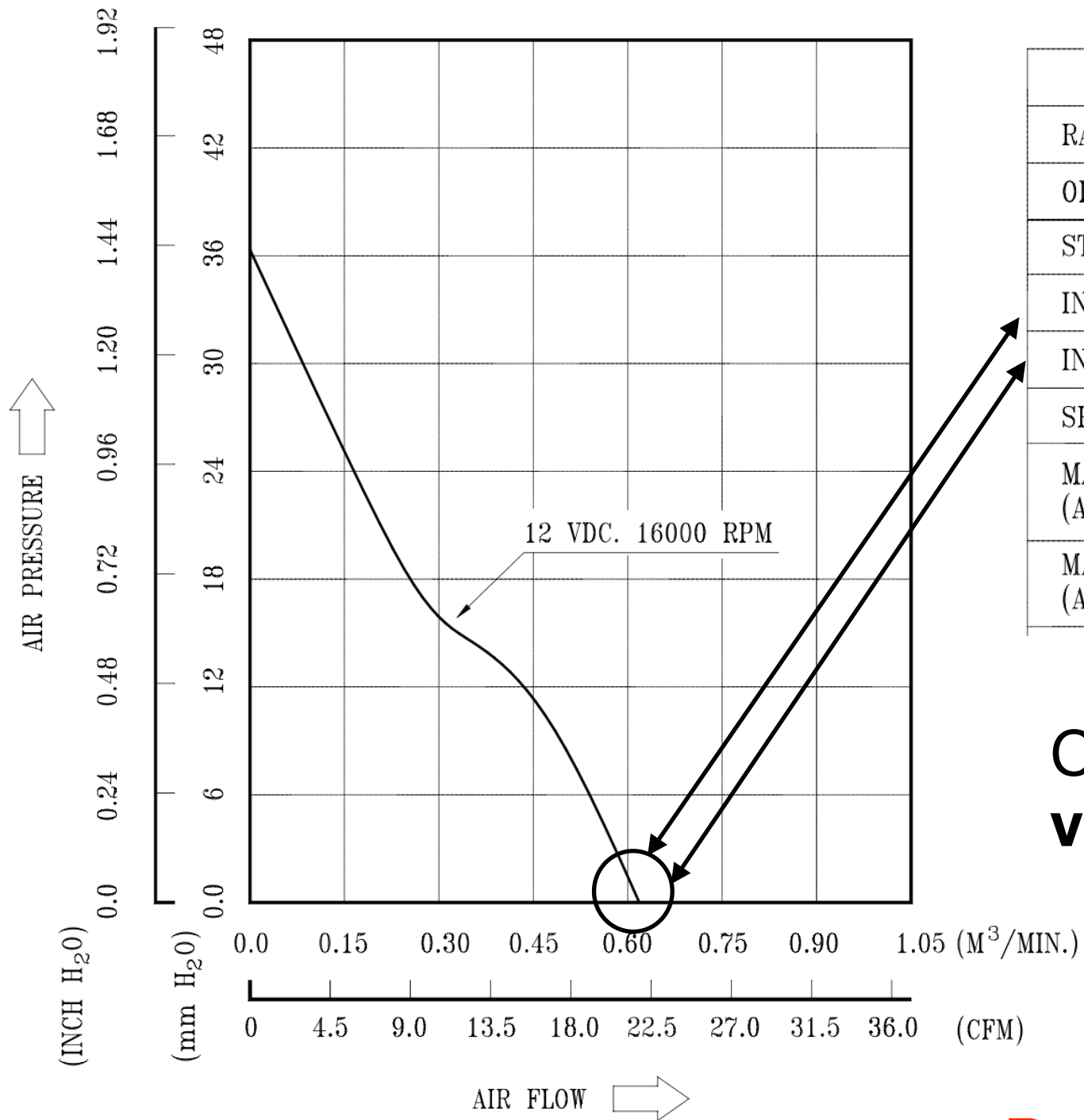
Mechanical power

$$P(V)Q(V)$$

going to zero in both
extreme scenarios

About current and power

Data sheet of PFB0412EHN-TP06



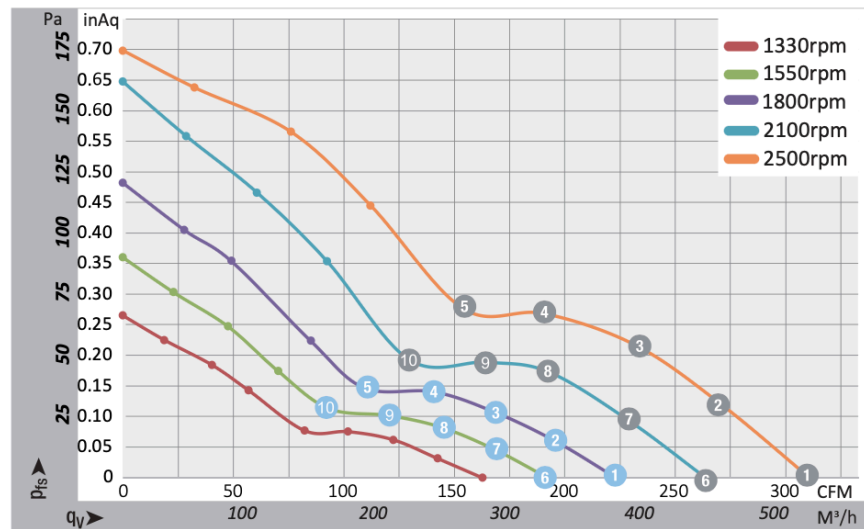
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Current and power given at **rated voltage and free-air conditions**

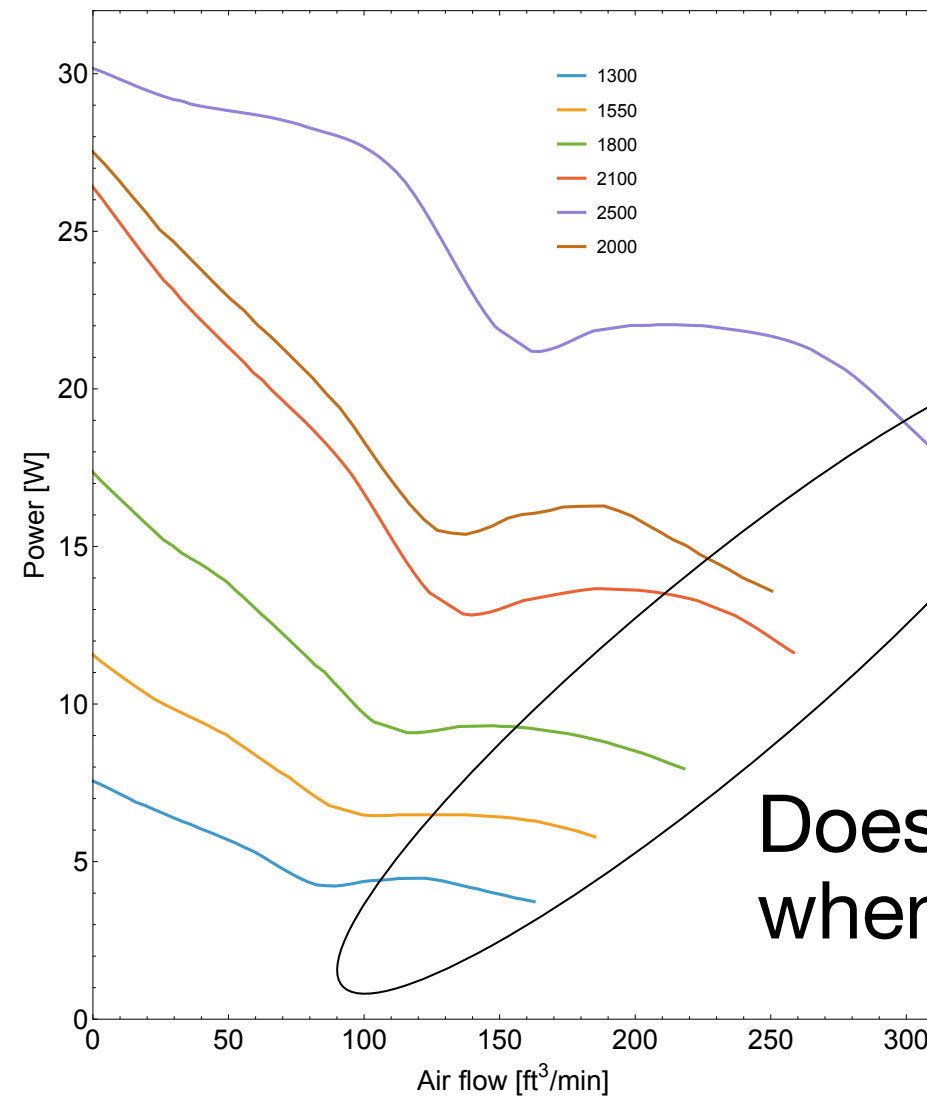
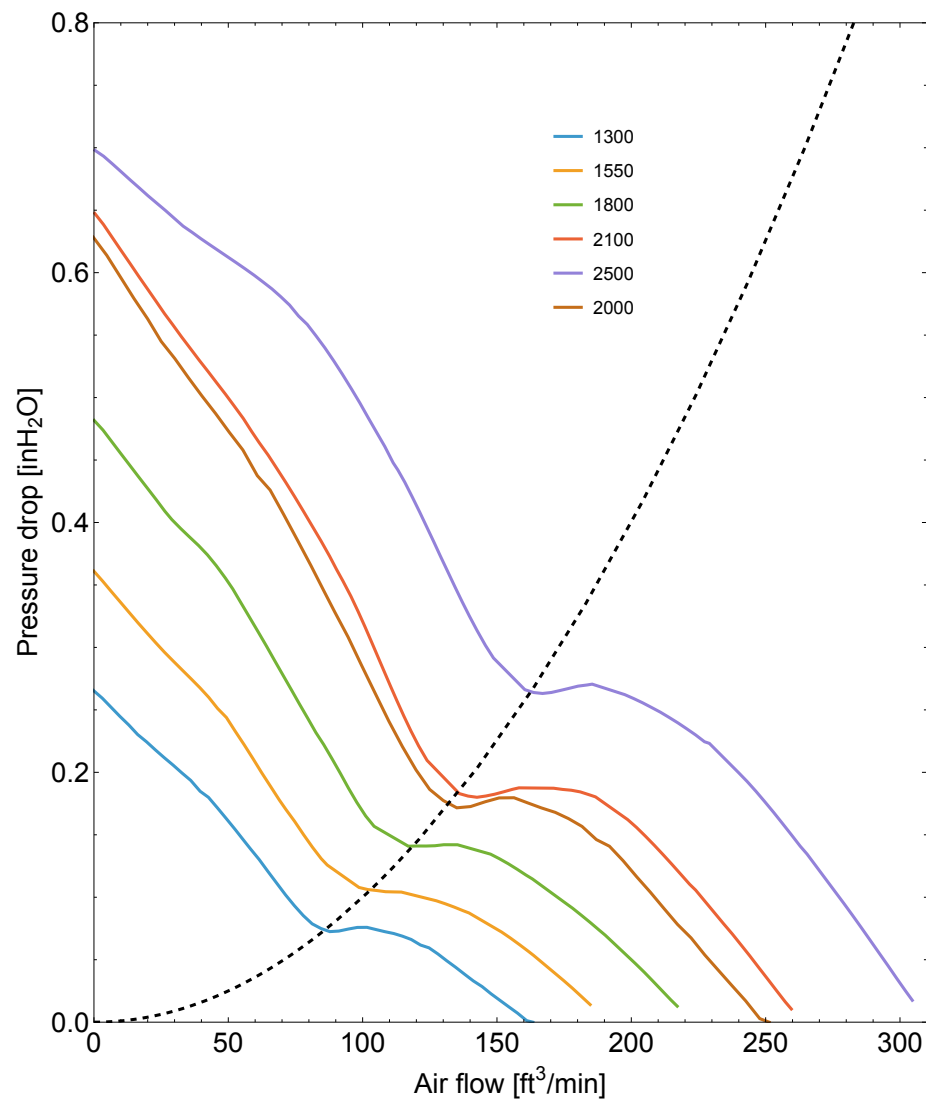
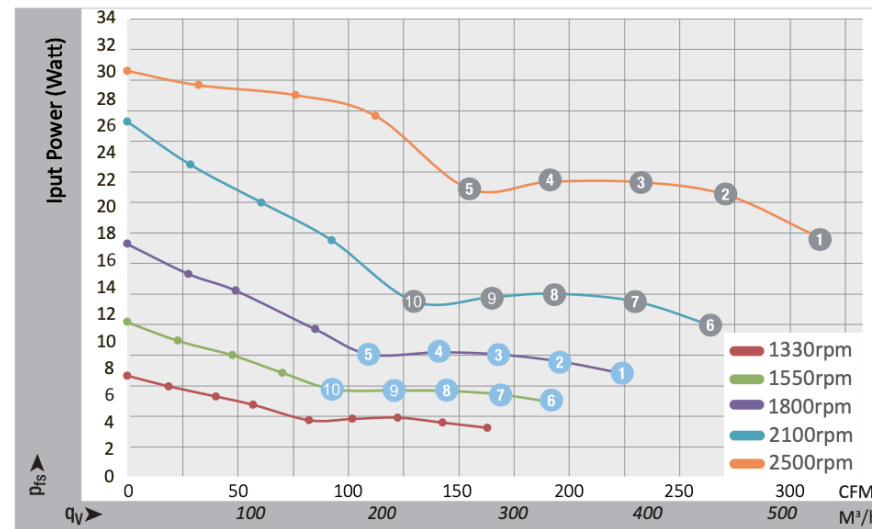
Does not replace a power-Q curve

Power-Q curve

Fan Curves



Power Curves

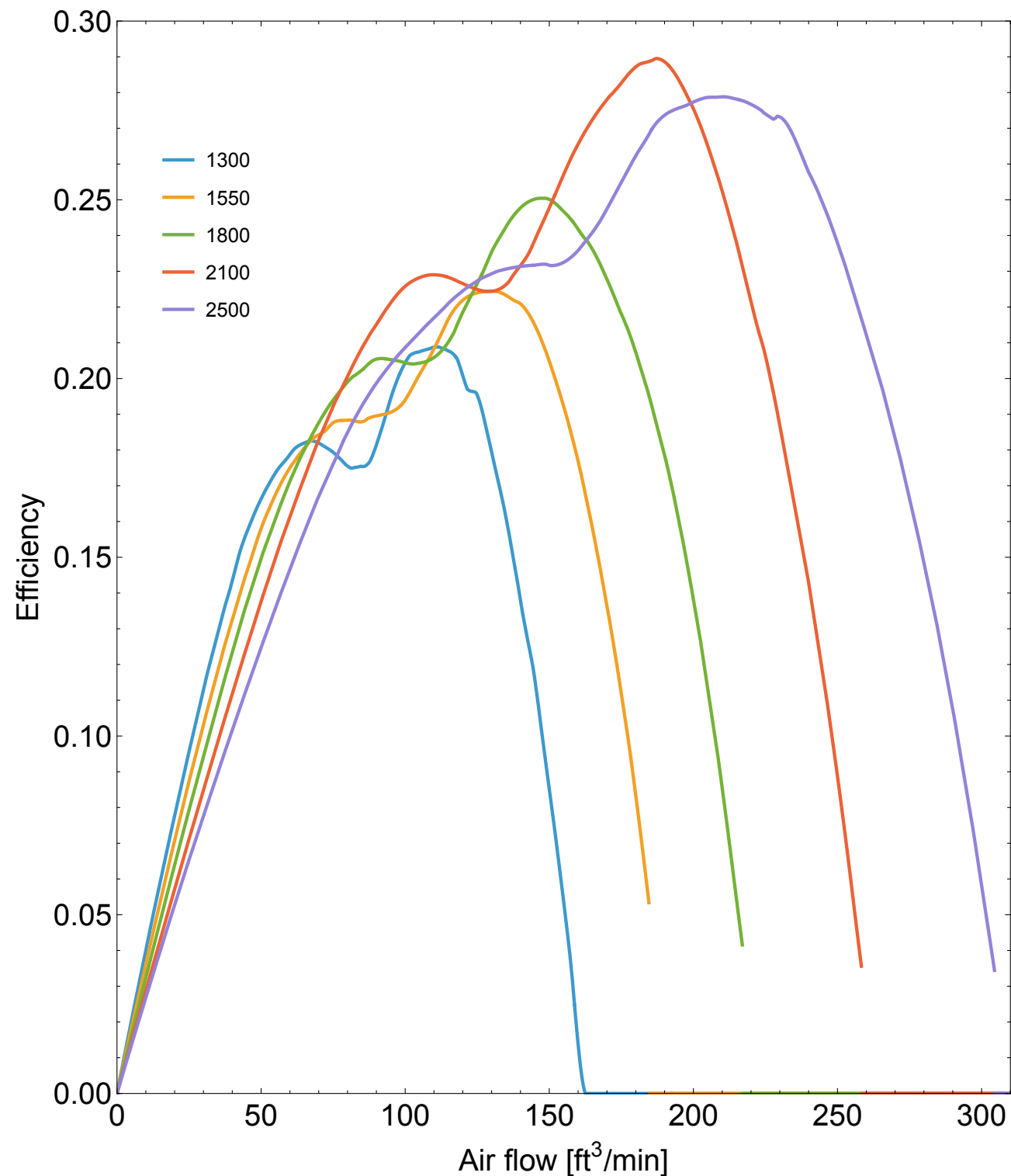


Does not go to 0
when P goes to 0

Efficiency-Q curve

Affinity laws predict a simple “stretch” for the efficiency curve

$$\eta = \frac{PQ}{HP}$$



Power-V curve

For a **fixed load**

the affinity laws predict a **cubic behavior**

$$\text{HP} = A(Z)Q^3 = B(Z)V^3 = C(Z)\text{RPM}^3$$

$A(Z), B(Z), C(Z)$

depend on Z

Cubic fit on Jacob's measurement

**Measurements done in
free-air conditions?**

